Exercise2: UNDERSTANDING DATA

Write Python programs to do the following operations:

1. Loading data from CSV file
2. Compute the basic statistics of given data - shape, no. of columns, mean
3. Splitting a data frame on values of categorical variables
4. Visualize data using Scatter plot

Dataset: brain_size.csv
Library: Pandas, matplotlib
a) Loading data from CSV file
b) Compute the basic statistics of given data - shape, no. of columns, mean
c) Splitting a data frame on values of categorical variables
d) Visualize data using Scatter plot

1. Loading data from CSV file

```
import pandas as pd
a=pd.read_csv("D:/data.csv")
print(a)
```

2. Compute the basic statistics of given data - shape, no. of columns, mean

```
import pandas as pd
a=pd.read_csv("D:/data.csv")
print('shape :',a.shape)
#no of columns
cols=len(a.axes[1])
print('no of columns:',cols)
#mean of data
m=a["marks"].mean()
print('mean of marks:',m)
```

3. Splitting a data frame on values of categorical variables

```
import pandas as pd
a=pd.read_csv("D:\data.csv")
```

```
print("Before:")
print(a)
a_split=a['address'].str.split(' ',1)
a['district']=a_split.str.get(0)
a['state']=a_split.str.get(1)
del(a['address'])
print("After:")
print(a)
```

4. **Visualize data using Scatter plot**

```
import pandas as pd
import matplotlib.pyplot as plt
a=pd.read_csv("D:\data.csv")
print("Before",a)
a_split=a['address'].str.split(',',1)
a['district']=a_split.str.get(0)
a['state']=a_split.str.get(1)
del(a['address'])
print("After=",a)
a.plot(kind='scatter',x='marks',y='rollno',c='red')
plt.show()
```